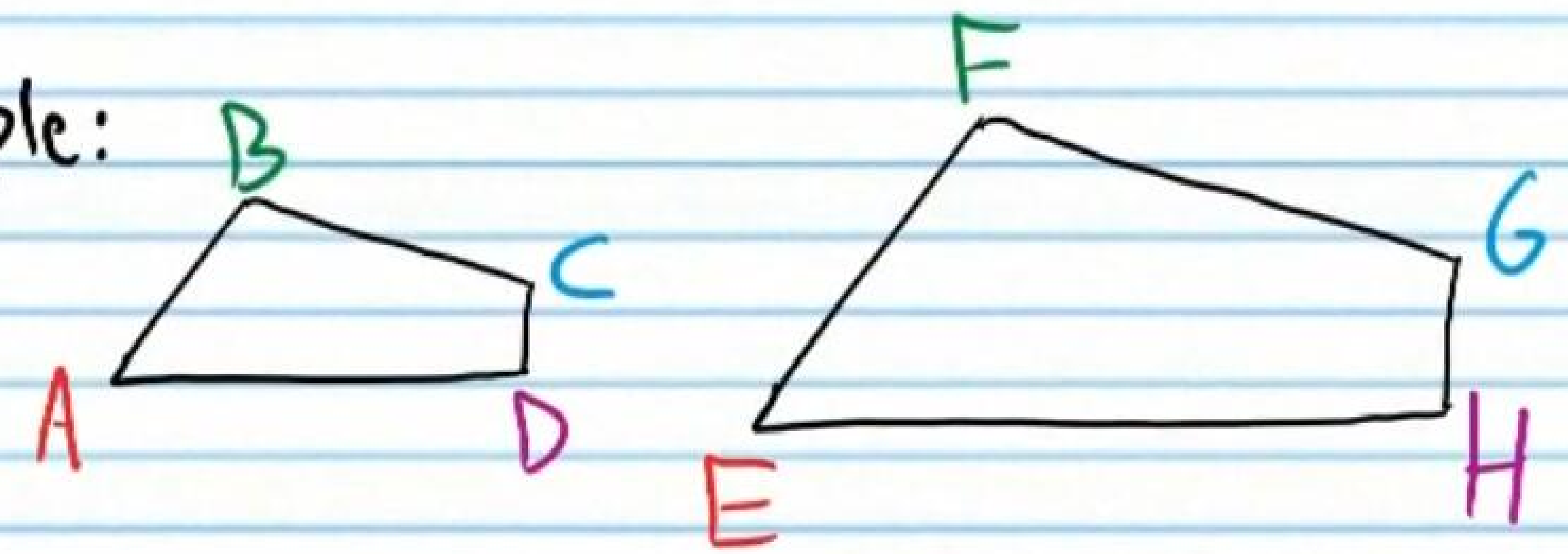
Name
Date
Course

Similar Polygons and Similarity Postulates and Theorems

Similar Polygons

Similar Polygons: Polygons with the same shape, but not always the same size. More specifically, polygons with congruent corresponding angles and equal ratios of corresponding sides (i.e. corresponding sides are proportional).

Example:

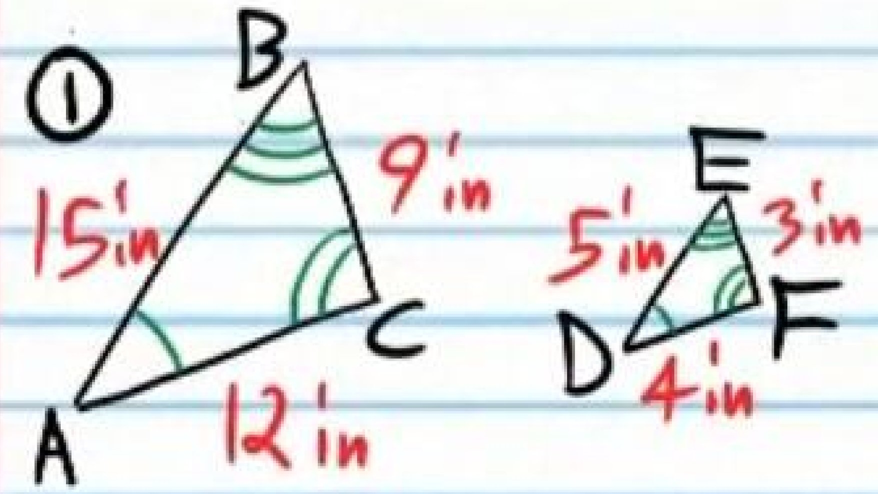


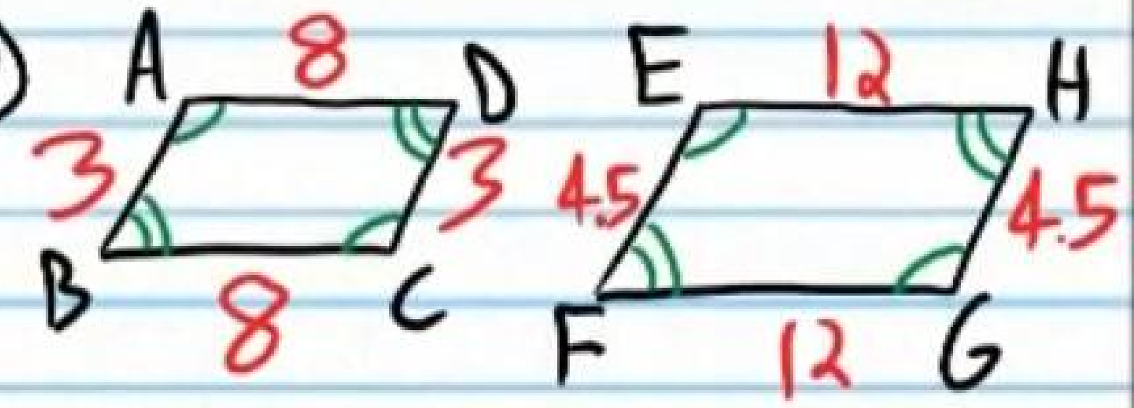
$\angle A \cong \angle E, \angle B \cong \angle F, \angle C \cong \angle G, \angle D \cong \angle H$

$$\frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE}$$

Similarity Statement: $ABCD \sim EFGH$

Show that the following polygons are similar. Then, write similarity statements.

① 

② 

①

I) $\angle A \cong \angle D, \angle B \cong \angle E, \angle C \cong \angle F$ ✓

II) $\frac{15}{5} = \frac{9}{3} = \frac{12}{4}$

$3 = 3 = 3$

$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$ ✓

III) $\Delta ABC \sim \Delta DEF$

②

I) $\angle A \cong \angle E, \angle D \cong \angle H, \angle B \cong \angle F, \angle C \cong \angle G$ ✓

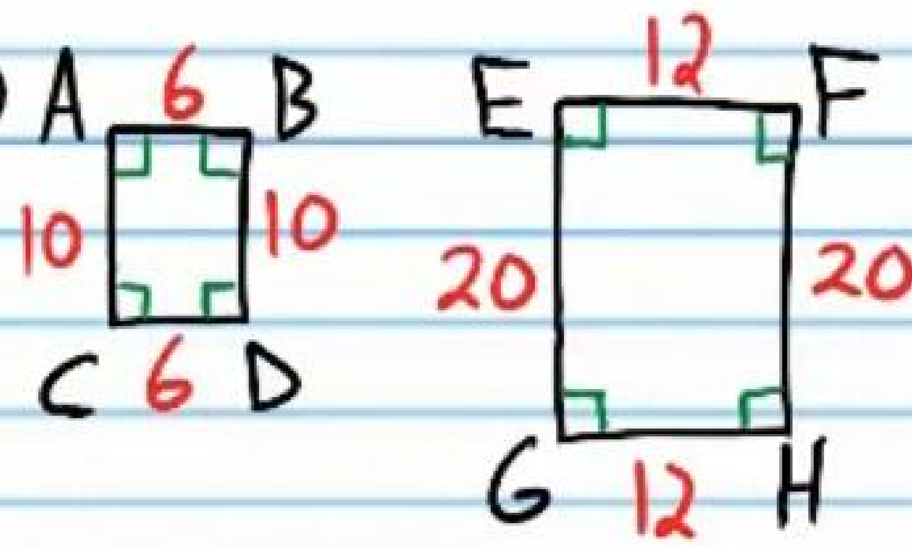
II) $\frac{8}{12} = \frac{8}{12} = \frac{3}{4.5} = \frac{3}{4.5}$

$0.\overline{6} = 0.\overline{6} = 0.\overline{6} = 0.\overline{6}$

$\frac{AD}{EH} = \frac{BC}{FG} = \frac{AB}{EF} = \frac{DC}{HG}$ ✓

III) $ABCD \sim EFGH$

③



I) $\angle A \cong \angle E$, $\angle B \cong \angle F$

$\angle C \cong \angle G$, $\angle D \cong \angle H$ ✓

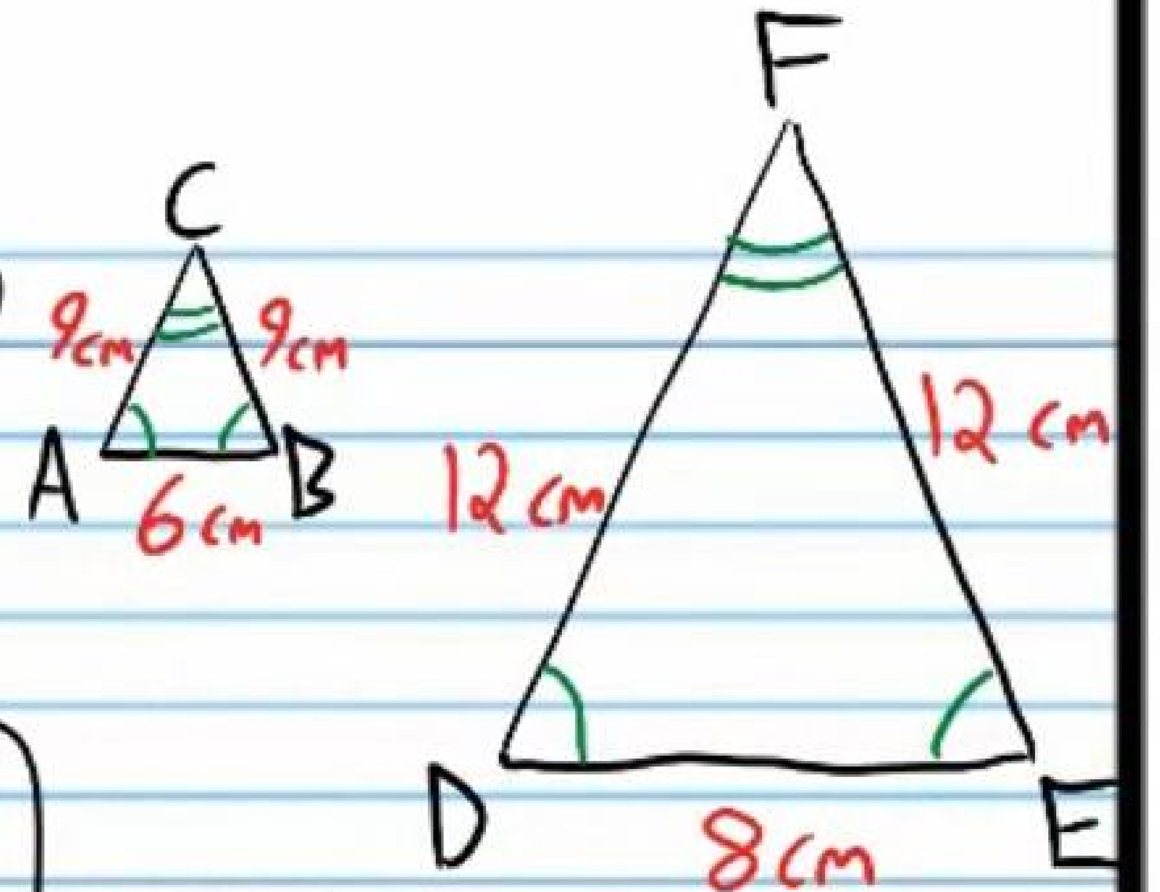
II) $\frac{6}{12} = \frac{6}{12} = \frac{10}{20} = \frac{10}{20}$

$\frac{1}{2} = \frac{1}{2} = \frac{1}{2} = \frac{1}{2}$

$\frac{AB}{EF} = \frac{CD}{GH} = \frac{AC}{EG} = \frac{BD}{FH}$ ✓

III) $ABDC \sim EFHG$

④



I) $\angle C \cong \angle F$, $\angle A \cong \angle D$

$\angle B \cong \angle E$ ✓

II) $\frac{9}{12} = \frac{9}{12} = \frac{6}{8}$

$\frac{3}{4} = \frac{3}{4} = \frac{3}{4}$

$\frac{AC}{DF} = \frac{CB}{FE} = \frac{AB}{DE}$ ✓

III) $\Delta ABC \sim \Delta DEF$

Scale Factor: The ratio of corresponding sides of similar polygons.

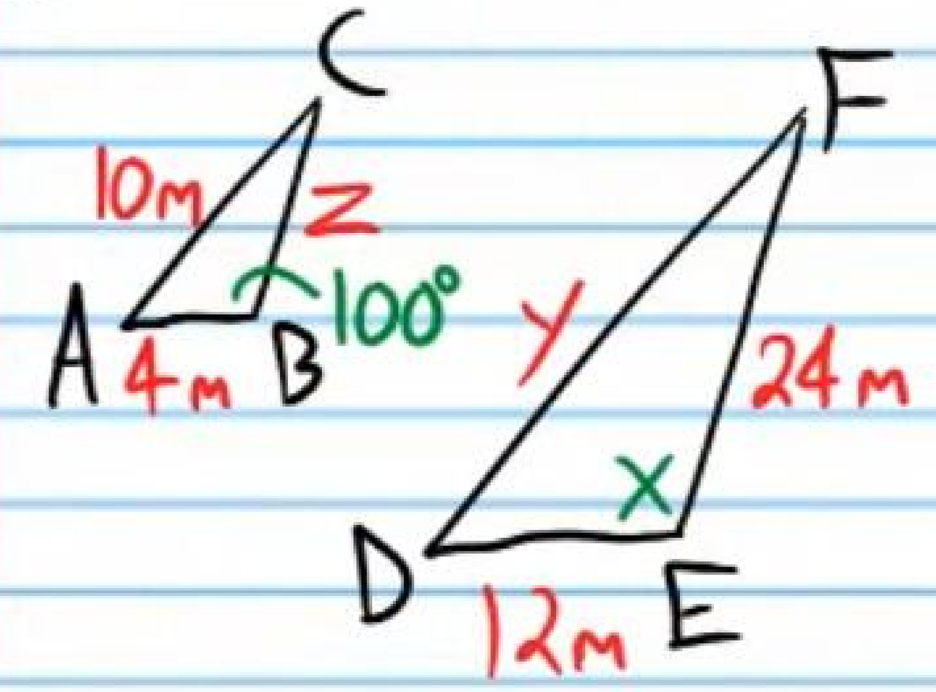
⑤ Find the scale factors for problem #'s 1-4.

I) 3 or $\frac{1}{3}$ II) $\frac{2}{3}$ or $\frac{3}{2}$ III) $\frac{1}{2}$ or $\frac{2}{1}$ IV) $\frac{3}{4}$ or $\frac{4}{3}$

$0.\bar{6}$ or 1.5

For each problem below, the polygons in each pair are similar. Find the scale factor and the unknowns.

⑥ $\Delta ABC \sim \Delta DEF$



I) $\boxed{x = 100°}$

II) $\frac{4}{12} = \frac{10}{y} = \frac{z}{24}$

$\frac{4}{12} = \frac{10}{y}$

$\frac{1}{3} = \frac{10}{y}$

$\boxed{y = 30m}$

$\frac{4}{12} = \frac{z}{24}$

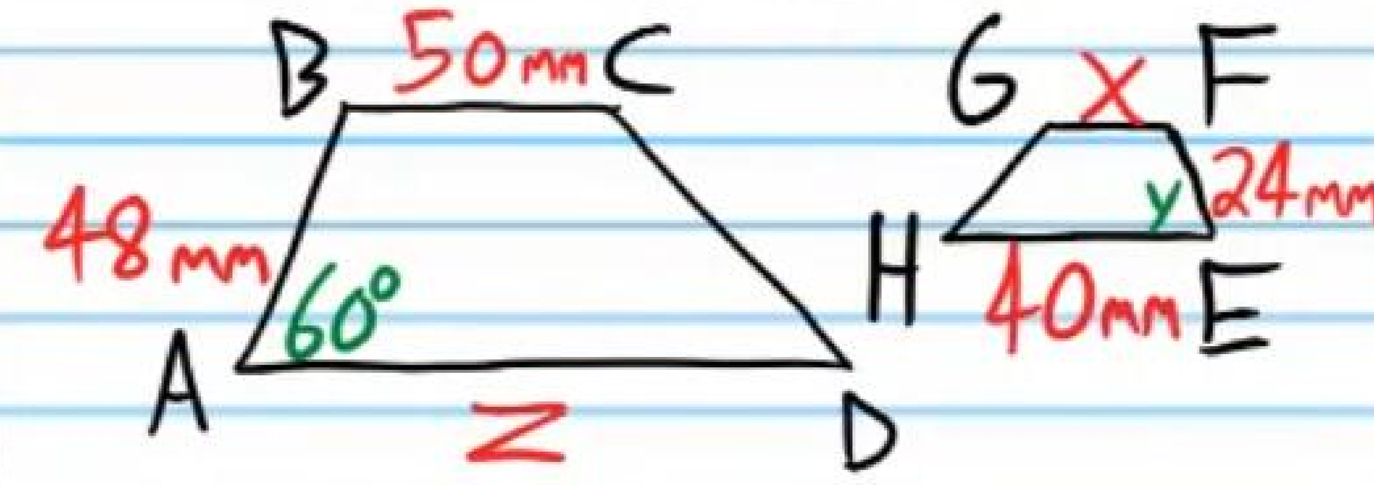
$\frac{1}{3} = \frac{z}{24}$

$24 = 3z$

$\boxed{8m = z}$

III) $\boxed{SF = \frac{1}{3} \text{ or } 3}$

⑦ $ABCD \sim EFGH$



I) $\boxed{y = 60°}$

II) $\frac{24}{48} = \frac{x}{50} = \frac{40}{z}$

$\frac{1}{2} = \frac{x}{50}$

$50 = 2x$

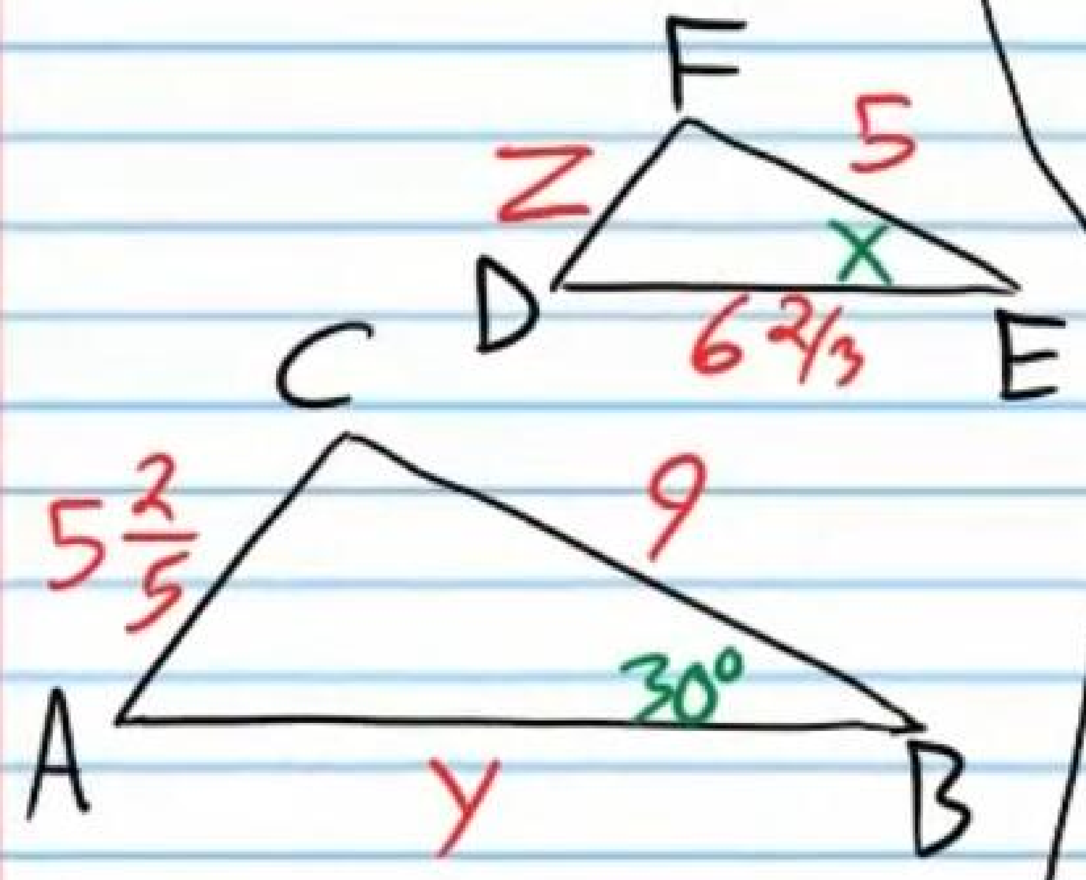
$\boxed{25mm = x}$

$\frac{1}{2} = \frac{40}{z}$

$\boxed{z = 80mm}$

III) $\boxed{SF = \frac{1}{2} \text{ or } 2}$

⑧ $\triangle ABC \sim \triangle DEF$



$\boxed{x = 30°}$

$$\frac{5}{9} = \frac{6\frac{2}{3}}{y} = \frac{z}{5\frac{2}{5}}$$

$\frac{5}{9} = \frac{6\,2/3}{y}$

$5y = 9 \cdot 6\frac{2}{3}$

$5y = \frac{9}{1} \cdot \frac{20}{3}$

$5y = \frac{180}{3}$

$5y = 60$

$\boxed{y = 12}$

$\frac{5}{9} = \frac{z}{5\,2/5}$

$9z = 5 \cdot 5\frac{2}{5}$

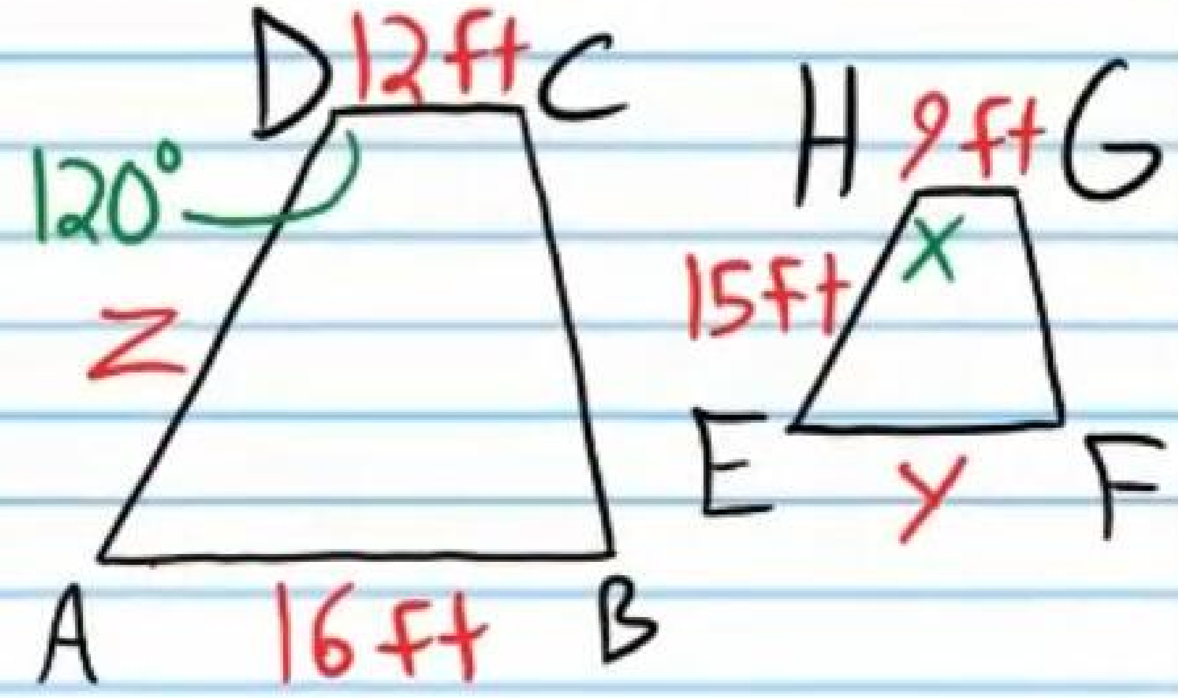
$= \frac{5}{1} \cdot \frac{27}{5}$

$9z = 27$

$\boxed{z = 3}$

$\boxed{SF = 5/9 \text{ or } 9/5}$

⑨ ABCD ~ EFGH



$\boxed{x = 120°}$

$$\frac{12}{9} = \frac{z}{15} = \frac{16}{y}$$

$\frac{12}{9} = \frac{z}{15}$

$\frac{4}{3} = \frac{z}{15}$

$3z = 60$

$\boxed{z = 20\text{ ft}}$

$\frac{12}{9} = \frac{16}{y}$

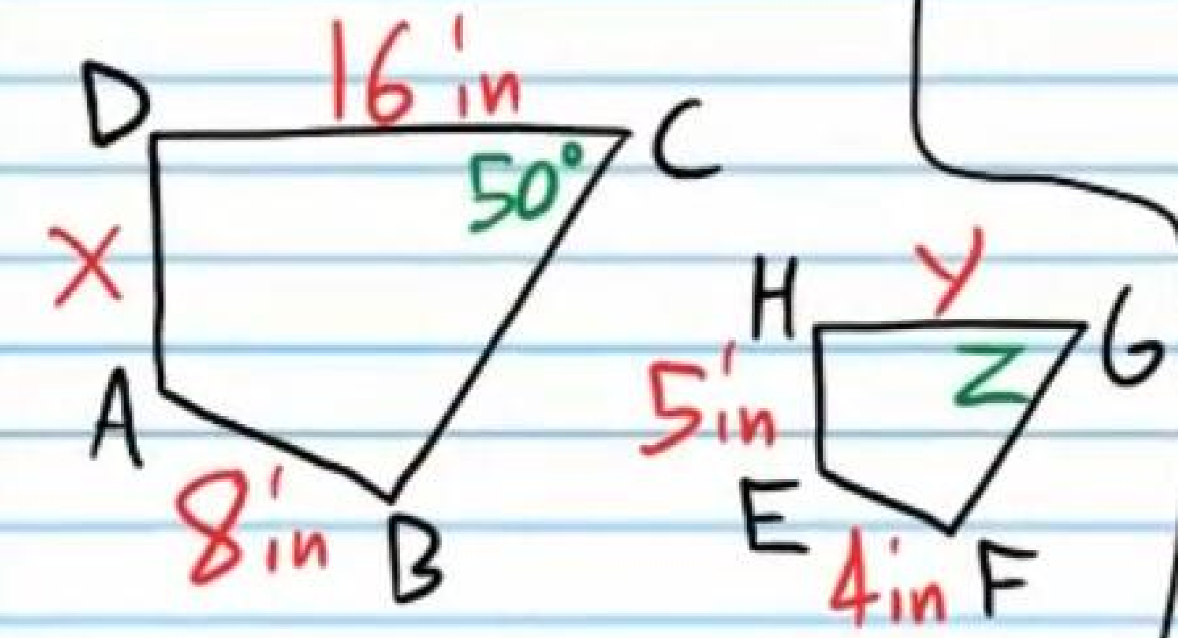
$\frac{4}{3} = \frac{16}{y}$

$4y = 48$

$\boxed{y = 12\text{ ft}}$

$\boxed{SF = \frac{4}{3} \text{ or } \frac{3}{4}}$

⑩ ABCD ~ EFGH



$\boxed{z = 50°}$

$$\frac{8}{4} = \frac{x}{5} = \frac{16}{y}$$

$\frac{2}{1} = \frac{x}{5}$

$\boxed{10\text{ in} = x}$

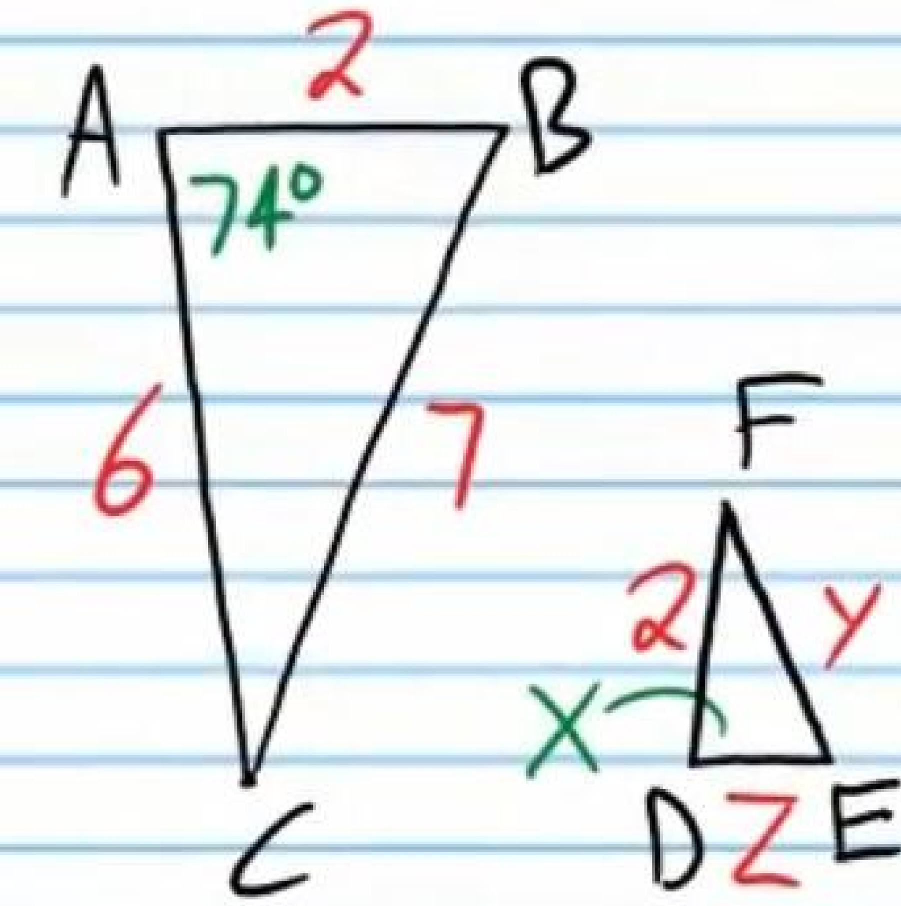
$\frac{2}{1} = \frac{16}{y}$

$2y = 16$

$\boxed{y = 8\text{ in}}$

$\boxed{SF = 2 \text{ or } 1/2}$

⑪ $\triangle ABC \sim \triangle DEF$



$\boxed{x = 74°}$

$$\frac{6}{2} = \frac{2}{z} = \frac{7}{y}$$

$\frac{3}{1} = \frac{2}{z}$

$3z = 2$

$\boxed{z = \frac{2}{3}}$

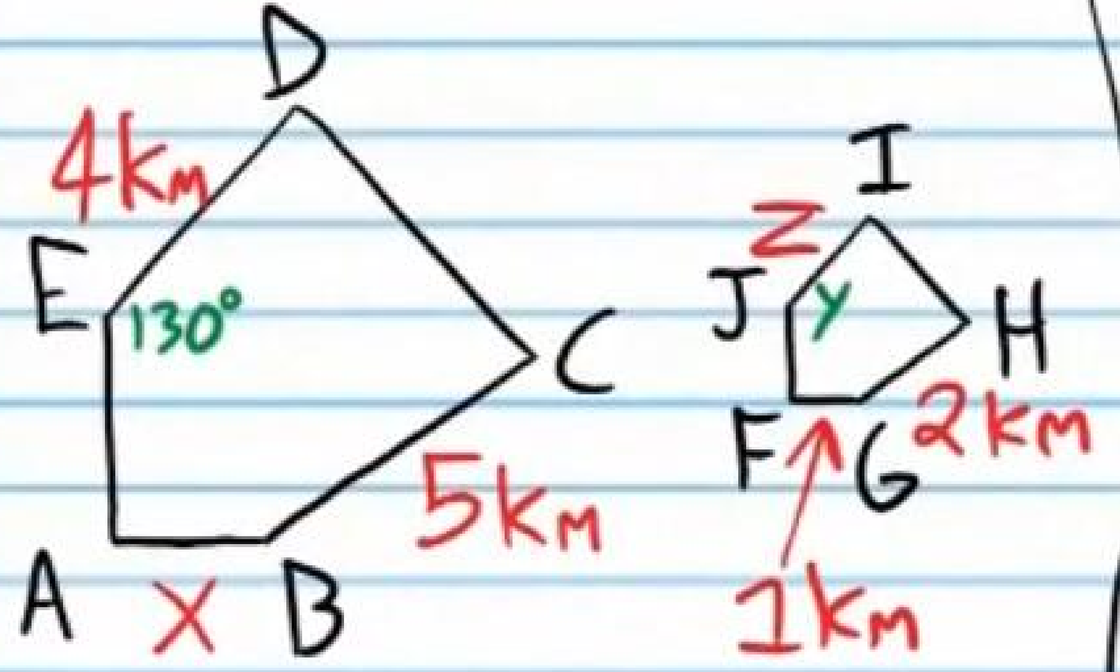
$\frac{3}{1} = \frac{7}{y}$

$3y = 7$

$\boxed{y = \frac{7}{3}}$

$\boxed{SF = 3 \text{ or } 1/3}$

(12) ABCDE ~ FGHIJ



$\boxed{y = 130°}$

$$\frac{5}{2} = \frac{4}{z} = \frac{x}{1}$$

$$\frac{5}{2} = \frac{4}{z}$$

$5z = 8$

$\boxed{z = \frac{8}{5}\text{km}}$

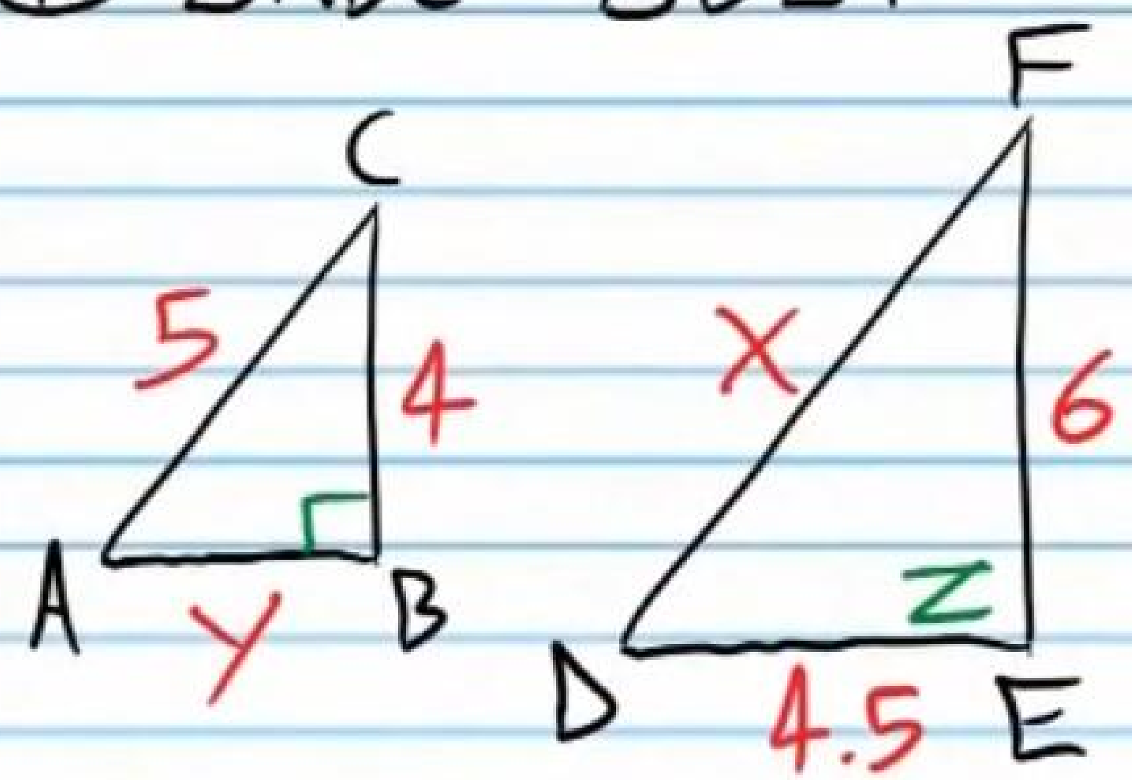
$$\frac{5}{2} = \frac{x}{1}$$

$5 = 2x$

$\boxed{5/2\text{ km} = x}$

$\boxed{SF = 5/2 \text{ or } 2/5}$

(13) $\triangle ABC \sim \triangle DEF$



$\boxed{z = 90°}$

$$\frac{4}{6} = \frac{5}{x} = \frac{y}{4.5}$$

$$\frac{2}{3} = \frac{5}{x}$$

$2x = 15$

$\boxed{x = 7.5}$

$$\frac{2}{3} = \frac{y}{4.5}$$

$9 = 3y$

$\boxed{3 = y}$

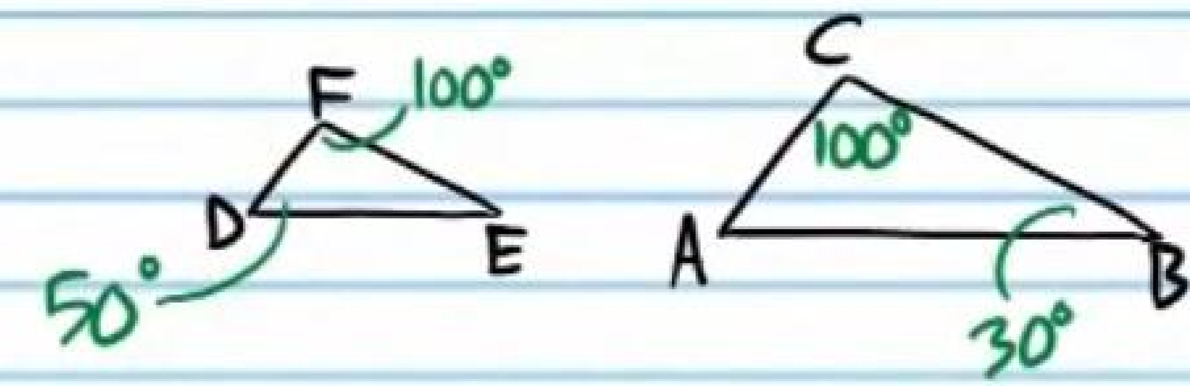
$\boxed{SF = \frac{2}{3} \text{ or } \frac{3}{2}}$

Similarity Postulates and Theorems

Angle-Angle Similarity Postulate: If two angles in one triangle are congruent to two angles in another triangle, then the triangles are similar. The postulate is abbreviated AA.

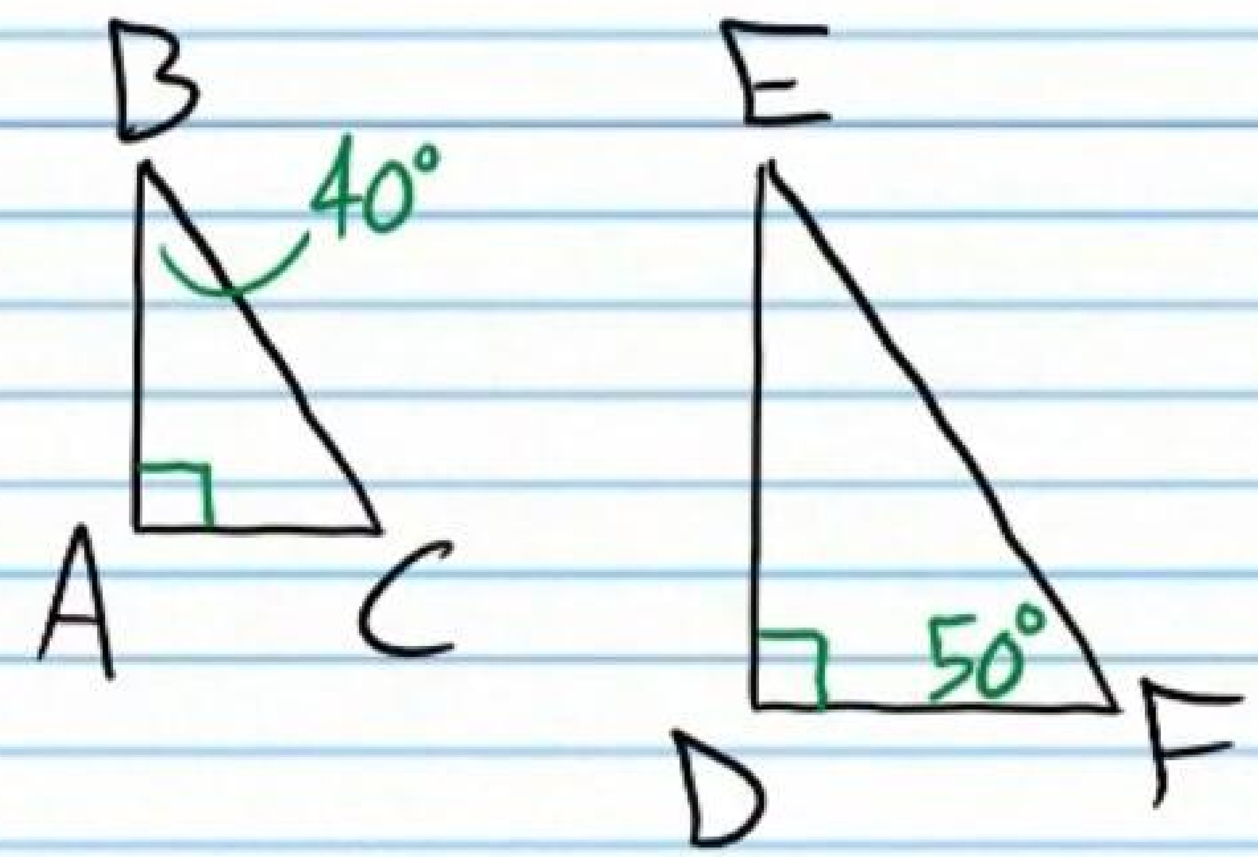
In the following problems, write two-column proofs showing that the triangles are similar.

(14)



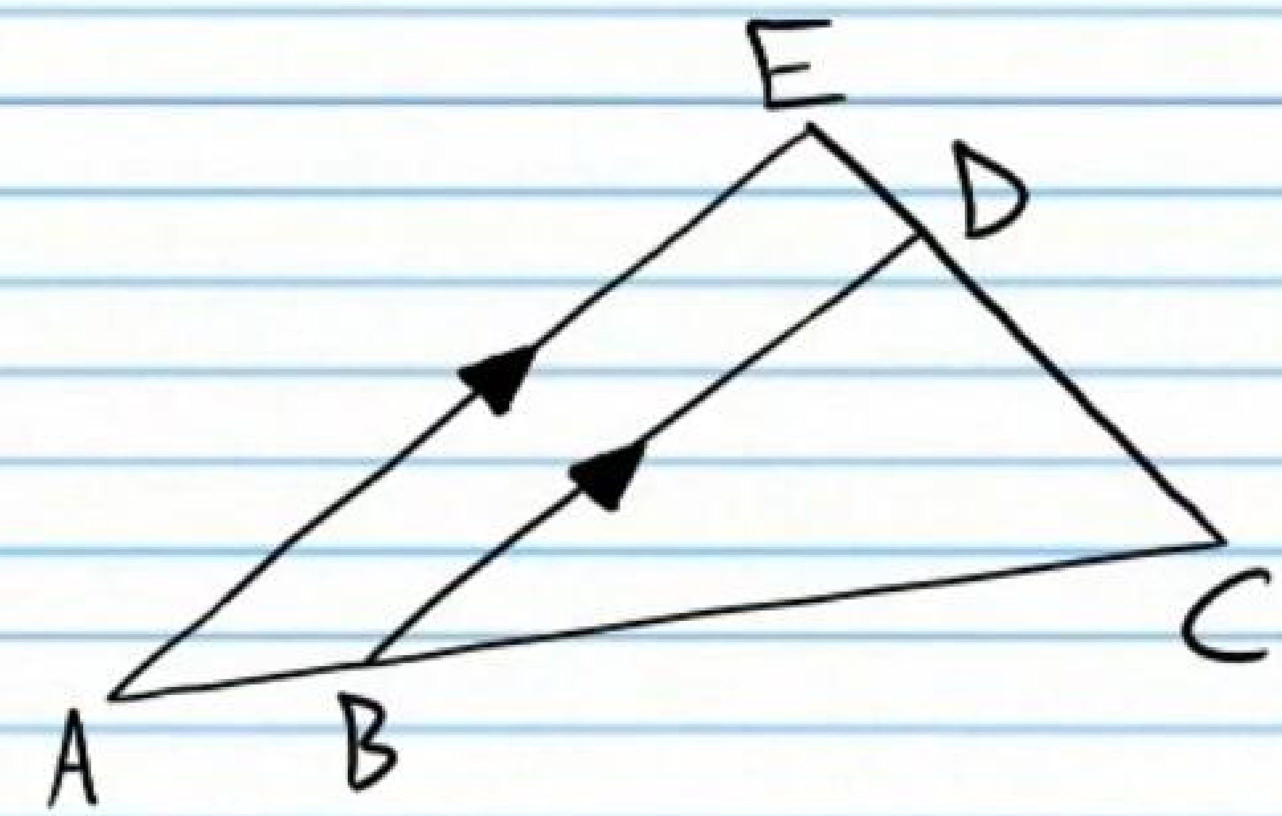
Statement	Reason
I) $m\angle D = 50°$, $m\angle F = 100°$, $m\angle C = 100°$, and $m\angle B = 30°$	Given (diagram)
II) $m\angle F + m\angle D + m\angle E = 180°$	$\triangle$ Sum Thm.
III) $100 + 50 + m\angle E = 180$	Subs. Prop of =.
IV) $150 + m\angle E = 180$	Simplification.
V) $m\angle E = 30°$	Subt. Prop of =.
VI) $m\angle E = m\angle B$, $m\angle C = m\angle F$	Trans. Prop of =.
VII) $\triangle DFE \sim \triangle ACB$	AA

(15)



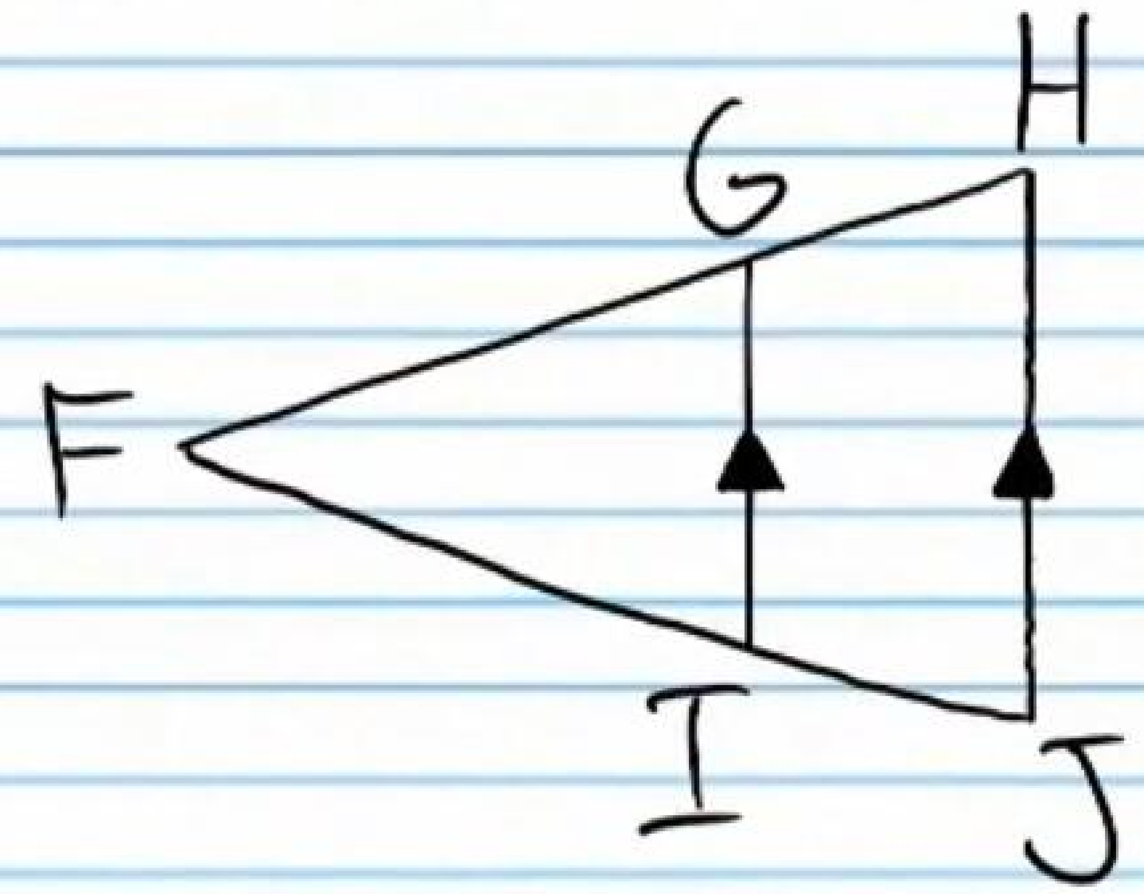
Statement	Reason
I) $m\angle B = 40°$, $m\angle A = 90°$, $m\angle F = 50°$, $m\angle D = 90°$	Given (diagram).
II) $m\angle A + m\angle B + m\angle C = 180°$	Δ Sum Thm.
III) $90 + 40 + m\angle C = 180$	Subs. Prop of =.
IV) $130 + m\angle C = 180$	simplification.
V) $m\angle C = 50°$	Subt. Prop of =.
VI) $m\angle A = m\angle D$, $m\angle C = m\angle F$	Trans. Prop of =.
VII) $\Delta ABC \sim \Delta DEF$	AA

(16)



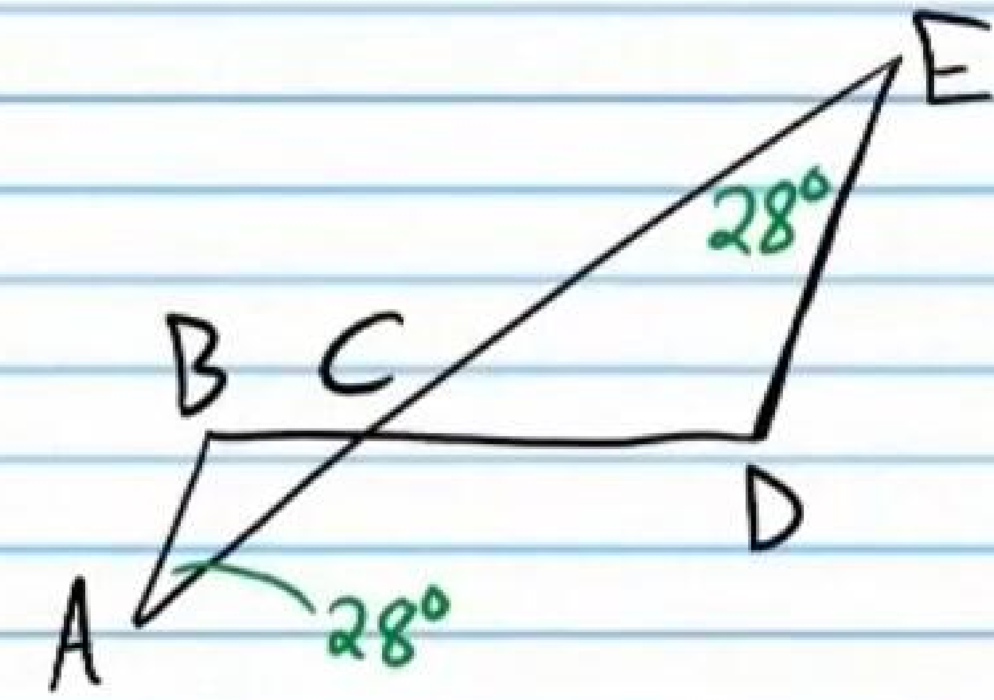
Statement	Reason
I) $\overline{AE} \parallel \overline{BD}$	Given (diagram)
II) $\angle EAC \cong \angle DBC$, $\angle AEC \cong \angle BDC$	CAP
III) $\Delta AEC \sim \Delta BDC$	AA

(17)



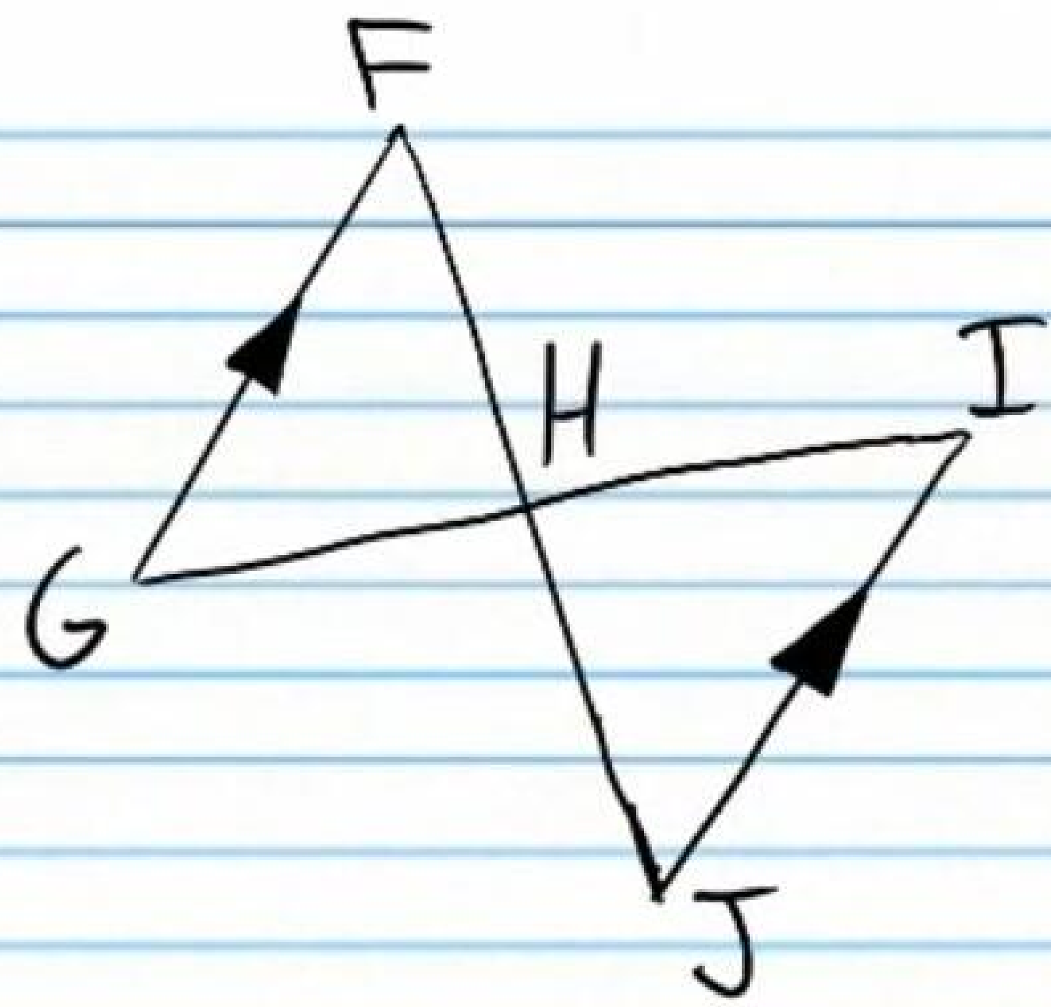
Statement	Reason
I) $\overline{GI} \parallel \overline{HJ}$	Given (diagram)
II) $\angle FGI \cong \angle FHJ$ and $\angle FIG \cong \angle FJH$	CAP
III) $\triangle FGI \sim \triangle FHJ$	AA

(18)



Statement	Reason
I) $m\angle E = 28°$ and $m\angle A = 28°$	Given (diagram)
II) $m\angle E = m\angle A$	Trans. Prop of =.
III) $m\angle BCA = m\angle ECD$	Vert. ∠'s Thm
IV) $\triangle ABC \sim \triangle EDC$	AA

(19)

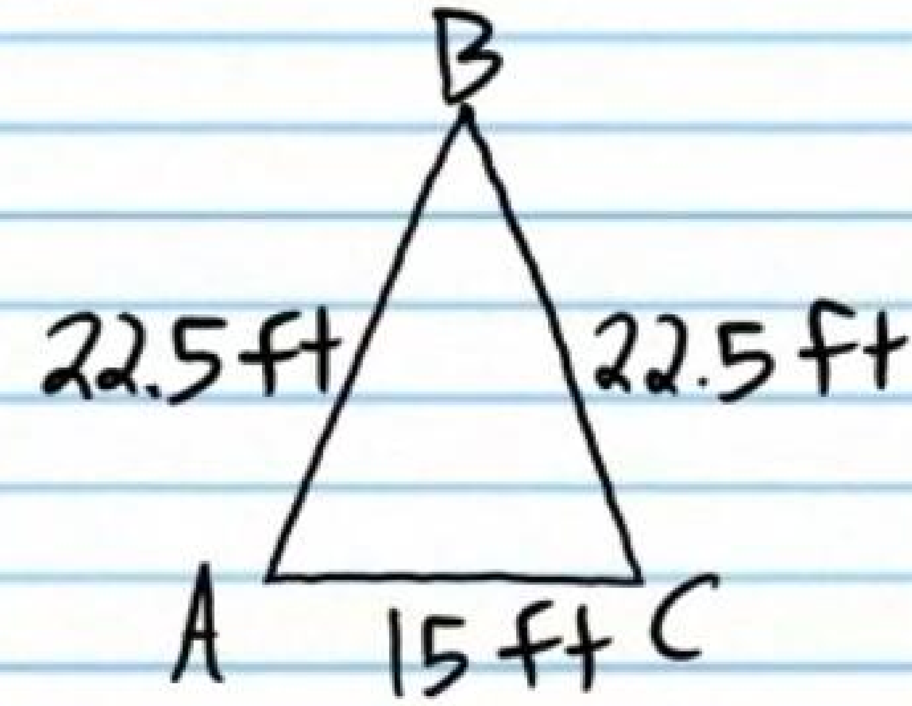


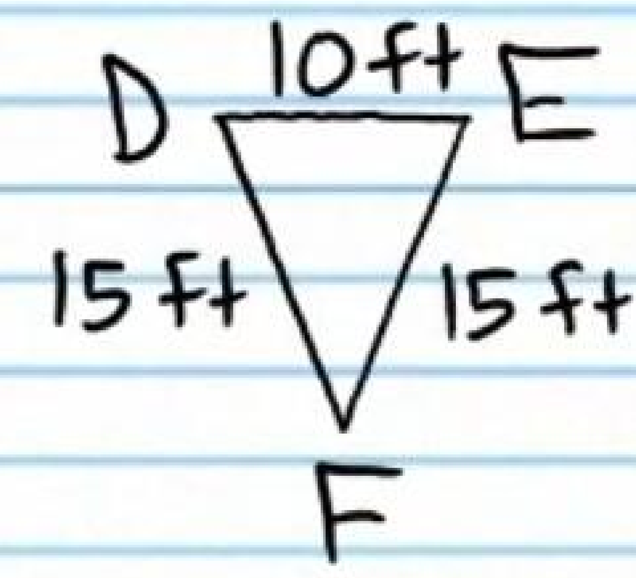
Statement	Reason
I) $\overline{GF} \parallel \overline{JI}$	Given (diagram)
II) $m\angle GFH = m\angle IJH$ $m\angle FGH = m\angle HIJ$	AIAT
III) $\triangle GFH \sim \triangle IJH$	AA

Side-Side-Side Similarity Theorem: If the ratios of corresponding side lengths of triangles are equal, then the triangles are similar. The theorem is abbreviated SSS.

Write two-column proofs to show that the triangles are similar.

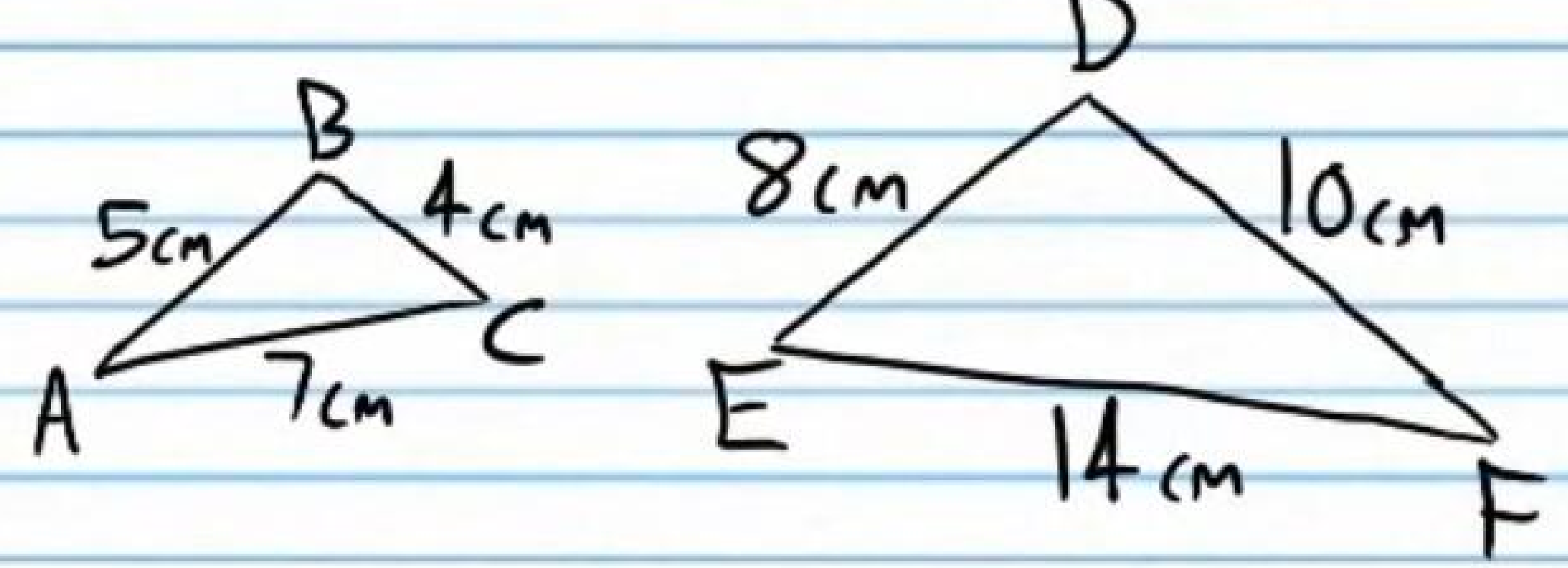
(20)





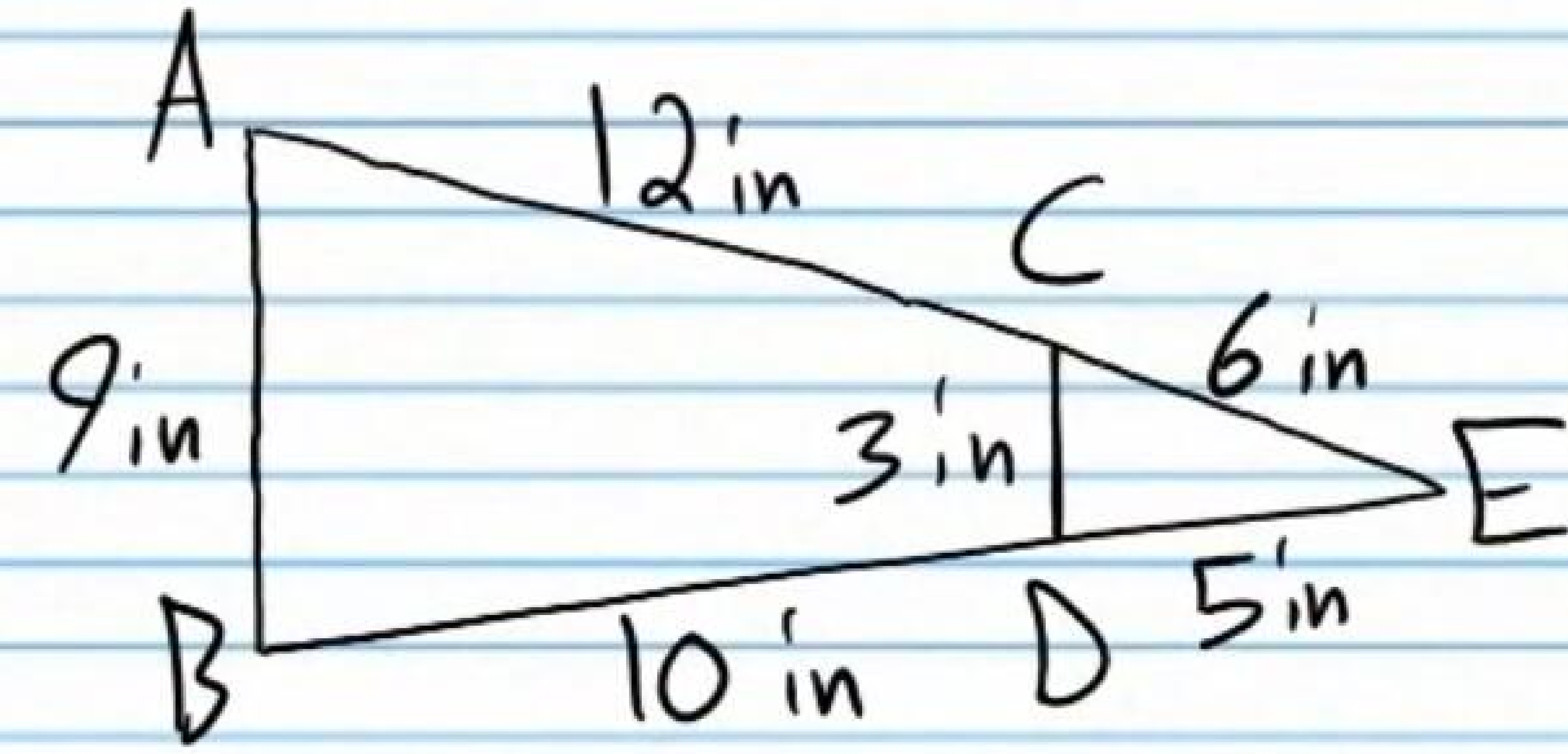
Statement	Reason
I) $AB = 22.5, BC = 22.5, AC = 15$ $DE = 10, DF = 15, EF = 15$	Given (diagram).
II) $\frac{AB}{DF} = \frac{22.5}{15}, \frac{BC}{FE} = \frac{22.5}{15}, \frac{AC}{DE} = \frac{15}{10}$	Subs. Prop of =.
III) $\frac{AB}{DF} = 1.5, \frac{BC}{FE} = 1.5, \frac{AC}{DE} = 1.5$	Simplification.
IV) $\frac{AB}{DF} = \frac{BC}{FE} = \frac{AC}{DE}$	Trans. Prop of =.
V) $\triangle ABC \sim \triangle DFE$	SSS

(21)



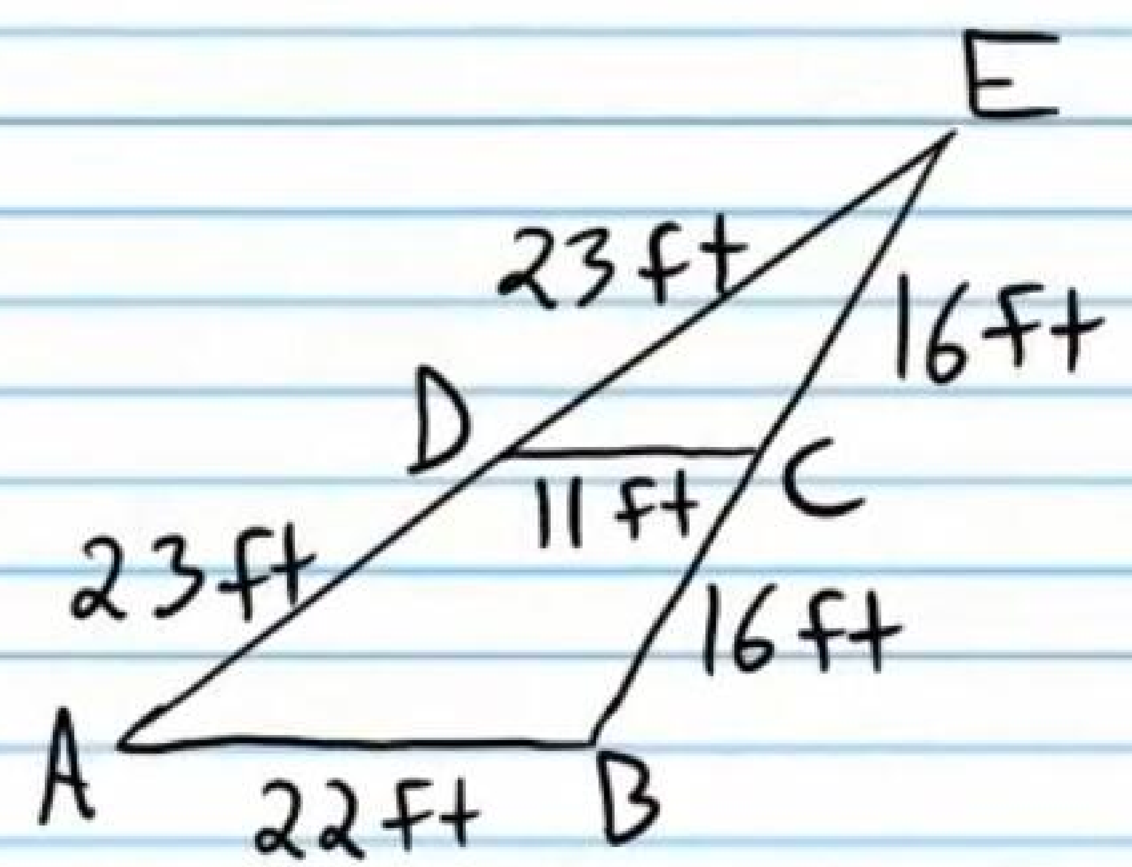
Statement	Reason
I) $AB = 5, BC = 4, AC = 7, ED = 8, DF = 10, EF = 14.$	Given (diagram).
II) $\frac{AC}{EF} = \frac{7}{14}, \frac{AB}{DF} = \frac{5}{10}, \frac{BC}{DE} = \frac{4}{8}.$	Subs. Prop of =.
III) $\frac{AC}{EF} = \frac{1}{2}, \frac{AB}{DF} = \frac{1}{2}, \frac{BC}{DE} = \frac{1}{2}$	simplification.
IV) $\frac{AC}{EF} = \frac{AB}{DF} = \frac{BC}{DE}$	Trans. Prop of =.
V) $\triangle ABC \sim \triangle FDE$	SSS

(22)



Statement	Reason
I) $AC = 12, CE = 6, AB = 9, BD = 10,$ and $DE = 5.$	Given (diagram).
II) $AE = AC + CE$ and $BE = BD + DE$	Seg Add Post.
III) $AE = 12 + 6$ and $BE = 10 + 5$	Subs. Prop of =.
IV) $AE = 18$ and $BE = 15$	simplification.
V) $\frac{AB}{CD} = \frac{9}{3}, \frac{AE}{CE} = \frac{18}{6}, \frac{BE}{DE} = \frac{15}{5}$	Subs. Prop of =.
VI) $\frac{AB}{CD} = 3, \frac{AE}{CE} = 3, \frac{BE}{DE} = 3$	simplification.
VII) $\frac{AB}{CD} = \frac{AE}{CE} = \frac{BE}{DE}$	Trans. Prop of =.
VIII) $\triangle ABE \sim \triangle CDE$	SSS

(23)

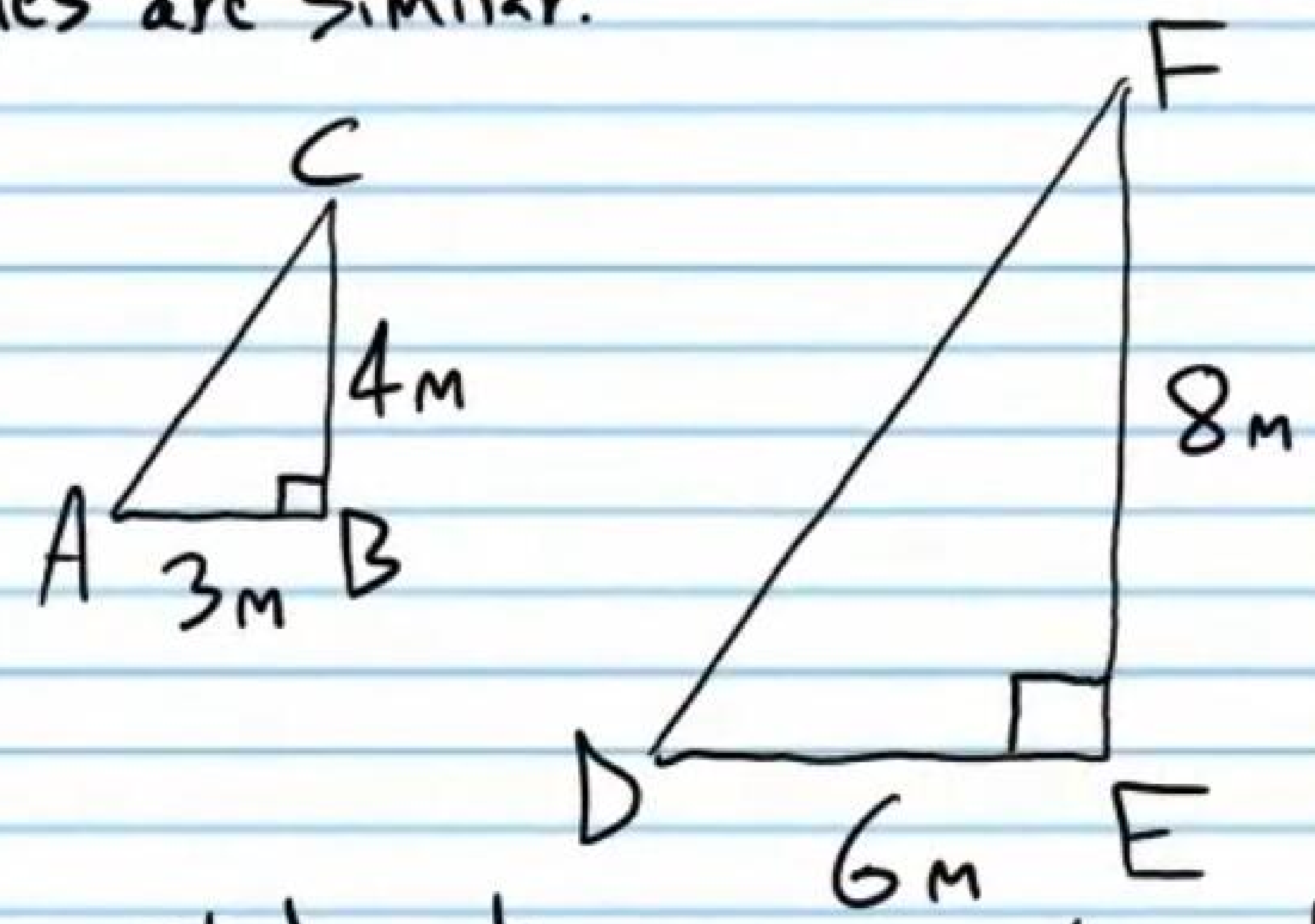


Statement	Reason
I) $AD = 23, DE = 23, AB = 22, BC = 16, CE = 16.$	Given (diagram).
II) $AE = AD + DE$ and $BE = BC + CE$	Seg Add Post.
III) $AE = 23 + 23$ and $BE = 16 + 16$	Subs. Prop of =.
IV) $AE = 46$ and $BE = 32$	Simplification.
V) $\frac{AE}{DE} = \frac{46}{23}, \frac{BE}{CE} = \frac{32}{16}, \frac{AB}{DC} = \frac{22}{11}$	Subs. Prop of =.
VI) $\frac{AE}{DE} = 2, \frac{BE}{CE} = 2, \frac{AB}{DC} = 2$	Simplification.
VII) $\frac{AE}{DE} = \frac{BE}{CE} = \frac{AB}{DC}$	Trans. Prop of =.
VIII) $\Delta ABE \sim \Delta DCE$	SSS

Side-Angle-Side Similarity Theorem: If the ratios of two corresponding sides of triangles are equal, and the angles between the sides in question are congruent, then the triangles are similar. The theorem is abbreviated SAS.

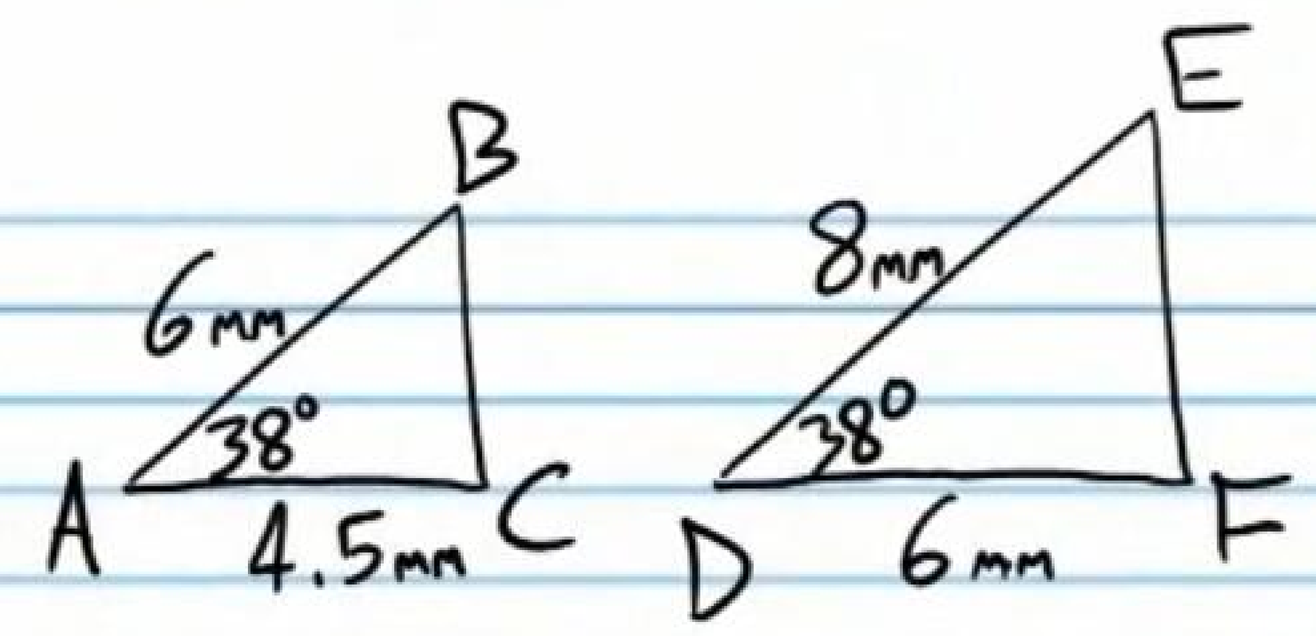
Write two-column proofs to show that the triangles are similar.

(24)



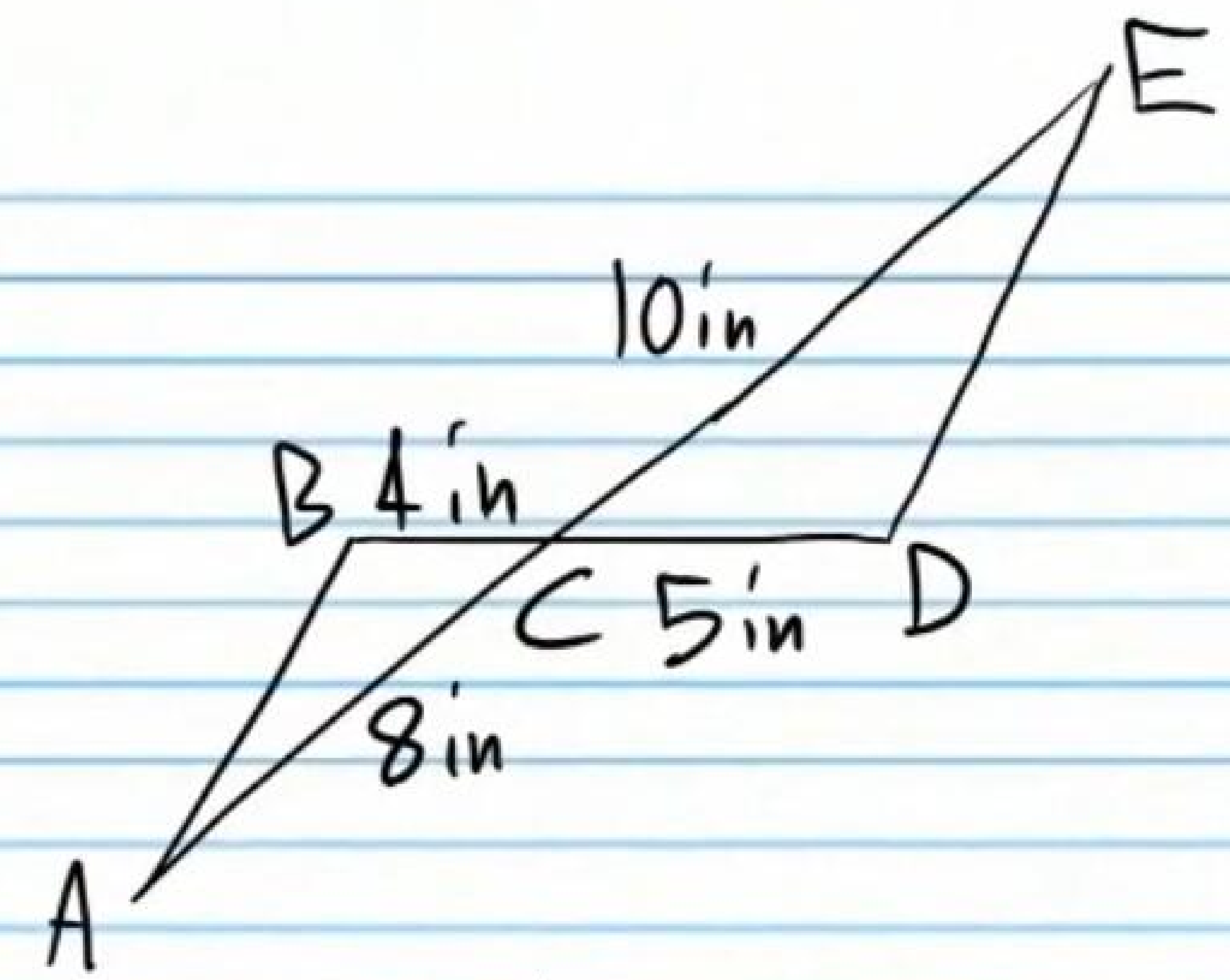
Statement	Reason
I) $AB = 3, BC = 4, m\angle B = 90°$ $DE = 6, FE = 8, m\angle E = 90°$	Given (diagram).
II) $m\angle B = m\angle E$	Trans. Prop of =.
III) $\frac{CB}{FE} = \frac{4}{8}$ and $\frac{AB}{DE} = \frac{3}{6}$	Subs. Prop of =.
IV) $\frac{CB}{FE} = \frac{1}{2}$ and $\frac{AB}{DE} = \frac{1}{2}$	Simplification.
V) $\frac{CB}{FE} = \frac{AB}{DE}$	Trans. Prop of =.
VI) $\Delta ABC \sim \Delta DEF$	SAS

(25)



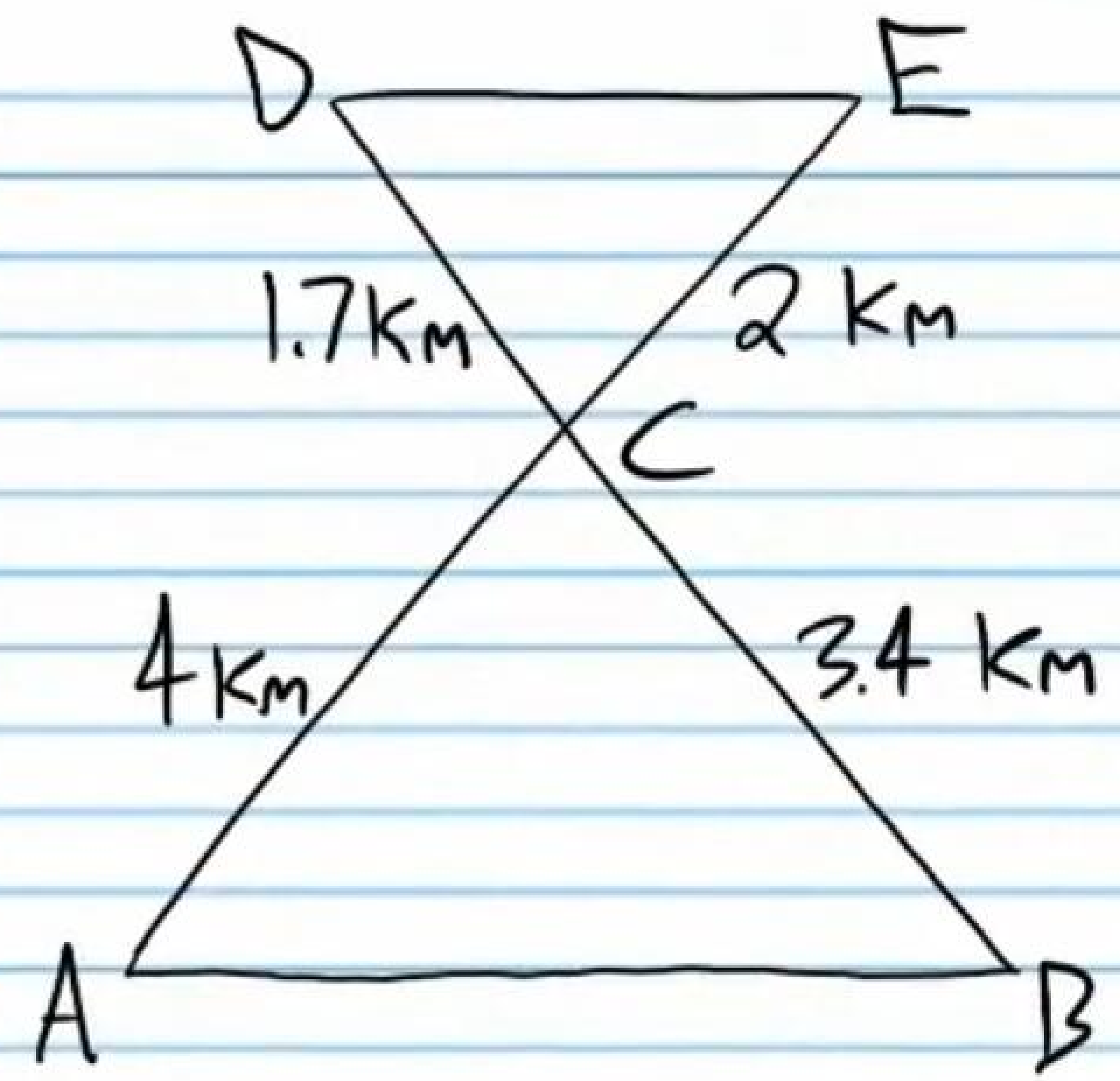
Statement	Reason
I) $AB=6, DE=8, AC=4.5, DF=6, m\angle A=38°, m\angle D=38°$	Given (diagram)
II) $m\angle A = m\angle D$	Trans. Prop of =.
III) $\frac{AB}{DE}=\frac{6}{8}$ and $\frac{AC}{DF}=\frac{4.5}{6}$	Subs. Prop of =.
IV) $\frac{AB}{DE}=0.75$ and $\frac{AC}{DF}=0.75$	Simplification
V) $\frac{AB}{DE}=\frac{AC}{DF}$	Trans. Prop of =.
VI) $\triangle ABC \sim \triangle DEF$	SAS

(26)



Statement	Reason
I) $BC=4, CD=5, AC=8, CE=10$	Given (diagram).
II) $m\angle BCA = m\angle ECD$	Vert. ∠'s Thm.
III) $\frac{BC}{CD}=\frac{4}{5}, \frac{AC}{CE}=\frac{8}{10}$	Subs. Prop of =.
IV) $\frac{BC}{CD}=\frac{4}{5}, \frac{AC}{CE}=\frac{4}{5}$	Simplification.
V) $\frac{BC}{CD}=\frac{AC}{CE}$	Trans. Prop of =.
VI) $\triangle ABC \sim \triangle EDC$	SAS

(27)



Statement	Reason
I) $DC=1.7, CE=2, AC=4, CB=3.4$	Given (diagram)
II) $m\angle DCE = m\angle ACB$	Vert. ∠'s Thm.
III) $\frac{AC}{CE}=\frac{4}{2}$ and $\frac{CB}{CD}=\frac{3.4}{1.7}$	Subs. Prop of =
IV) $\frac{AC}{CE}=2$ and $\frac{CB}{CD}=2$	Simplification
V) $\frac{AC}{CE}=\frac{CB}{CD}$	Trans. Prop of =
VI) $\triangle ABC \sim \triangle EDC$	SAS